Arin Nigam

Software Engineer - AI & Backend — Multi-agent systems and distributed backends

arin.nigam@gmail.com — +91 7042052330 — linkedin.com/in/arin-nigam — arinnigam.vercel.app — github.com/ArinNigam

Experience

Software Engineer, Cityflo Jul 2025 – Present
Mumbai, India

- **Komo AI:** Pitched in a **6-hour hackathon**, shipped to production a **multi-agent investigation platform** (Python, FastAPI, Claude Agent SDK) for debugging production incidents and customer issues. Used by **80+** engineers, ops, and founders; cut investigation time from **30–60 minutes** to **2–3 minutes** across **30–40 daily investigations.**
- Designed Komo's underlying **agent framework**: declarative agent specs, per-agent **MCP tool servers**, hard budget caps per run, and **inter-agent handoff** for cross-domain queries. Extended it to ship two products - an engineering **incident-triage agent** (**17 tools** spanning observability + code) and a **customer intelligence agent** (**11 tools** spanning analytics + CRM).
- **Sage AI:** Core contributor to Cityflo's **AI customer support platform** (Go, Postgres, Redis, React) handling **36K weekly chats** at **80%** AI deflection - cut human-handled contacts per 1K rides by **56%** within a month. Built the AI-human handoff orchestration, a priority queue for concurrent conversation routing, and the **internal support dashboard.**
- Built **SageAI's per-room concurrency** layer for stateful **real-time** conversations (LiveKit) across **horizontally-scaled ECS**, eliminating a race condition where multiple instances could spawn duplicate agents for the same room: **goroutine-per-room** with buffered channels, **Redis SetNX** for exactly-once agent spawn, and **Redis Pub/Sub** for cross-instance command delivery.

Software Engineer Intern, Walmart Global Tech Jan 2025 – Jun 2025
Bengaluru, India

- Shipped **12+ admin portal APIs** (Spring Boot, GraphQL, MongoDB) for Confido - Walmart's **seller-risk governance platform**, replacing manual DB/Postman operations previously used by **40+** risk analysts with **RBAC-gated workflows** for seller enforcement, offer publishing, and attribute onboarding. **Integrated SSO** and dual-admin approval flows.
- Built an **automated regression framework** (Airflow DAGs, BigQuery validation, Kafka consumers, GCS-hosted reports) covering **5 production risk strategies** (fulfillment SLA, GMV limits, chargebacks, refund disputes, customer contacts) - cut manual validation cycles from **6–8 hours** to under **30 minutes**, enabling daily runs that previously ran weekly.

Projects

Your Health Copilot [Dashboard] [Demo] 2025

- Full-stack **AI health assistant** helping patients understand their own **medical records** - upload prescriptions, lab reports, and imaging, then query them in natural language. Live with **100+** users. Stack: Next.js, LangChain, Supabase, AWS.
- Built a **multi-document RAG pipeline** with medical-context-aware chunking (respecting report boundaries and terminology) and **hybrid retrieval** (vector + keyword) - enabling **cross-report queries** like "*how has my HbA1c trended over 3 years*" across a user's and family's **medical history.**
- Cut query latency by **40%** via **response caching** for repeated queries, model routing (Haiku for lookups, Sonnet for synthesis), and **retry logic** with exponential backoff. Implemented **conversation memory** to persist context across sessions.

Smart Insti App [Video] 2024

- **Academic super-app** for IIT Bhilai (Flutter, Node.js, MongoDB) built via **OpenLake's Winter of Code** - a unified mess menu, lost-and-found, and campus broadcasts that were previously fragmented across WhatsApp groups. Adopted by **200+ DAUs** among students, faculty, and administrators. **Winner**, AutomateX Hackathon 2024.
- Architected **12+** REST endpoints with **RBAC** across three user roles (student, faculty, admin) and a customizable dashboard where users could pin frequently-used features. Cut server **p95 latency** by **25%** via query optimization and Redis caching.

Technical Skills

AI: Claude Agent SDK, MCP, RAG, multi-agent orchestration, LangChain, LLM evals, vector embeddings, prompt engineering

Languages: Go, Python, Java, TypeScript, C++

Backend: FastAPI, Gin, Spring Boot, Next.js, Node.js, REST, GraphQL, WebSocket

Distributed Systems: horizontally-scaled services, real-time streaming (WebRTC, LiveKit), Redis coordination, message queues (Kafka, SQS), Pub/Sub patterns

Data & Storage: PostgreSQL, Redis, ClickHouse, MongoDB, Aurora, BigQuery

Cloud/DevOps: AWS (ECS, EC2, S3, SQS), GCP (BigQuery, GCS), Docker, CI/CD, Datadog, Sentry, Airflow

Education

Birla Institute of Technology and Science (BITS), Pilani Jul 2021 – Jul 2025
B.E. Electrical and Electronics Engineering CGPA: 8.1/10.0

Achievements

- **AutomateX Hackathon 2024** – Winner, for Smart Insti App
- **LeetCode** – 1000+ problems solved